

Course: Operating system

Subject code : 24BTDS145

KrishiRakshak: AI-Based Early Detection and Management of Crop Diseases & Pest Infestations

Member PRN No	Name
24BTDSXXX	[Member name]
24BTDSXXX	[Member name]
24BTDSXXX	[Member name]
24BTDSXXX	[Member name]

Context and background

- Plant pests and diseases destroy up to 40% of global crop production every year (FAO).
- Farmers usually notice a disease only after visible damage has spread across the field.
- Extension officers cover very large areas; lab diagnosis and expert advice are slow to reach villages.
- Weather, crop stage, variety, soil and local pest history drive outbreaks — but are rarely combined into farm-level alerts.
- Wrong or late diagnosis leads to excess pesticide use, higher cost, residues and yield loss.

40%

of crop production lost to pests & diseases
(FAO)

38

crop disease & health classes recognised
by our AI

6

languages: English, हिन्दी, తెలుగు, தமிழ்,
ಕನ್ನಡ, मराठी

Early detection and management of crop diseases and pest infestations

Why is this project necessary?

Late detection

Symptoms are recognised only after the damage spreads.

Limited expert reach

Few extension staff and labs for many farmers.

Scattered risk signals

Weather, crop stage, soil and pest history are not combined.

Wrong treatment

Misdiagnosis → excess or wrong pesticide, cost, residues.

No surveillance picture

Officials lack real-time outbreak maps and data.

Language barrier

Advice is rarely available in the farmer's language.

Objectives and Key Goals

1. Detect

Identify diseases from a leaf photo with a CNN model (>95% target accuracy).

2. Forecast

7-day weather-based disease & pest risk for each farm.

3. Monitor

Pest-trap counts and IoT sensors with economic-threshold (ETL) alerts.

4. Map

Automatic geospatial hotspot detection and alerts to nearby farms.

5. Validate

Expert confirmation / correction of every AI diagnosis; lab referral.

6. Advise

Integrated Pest Management (IPM) advice with safe pesticide use, in 6 languages with audio.

7. Follow up

3/7-day follow-up checks; escalate cases that are not improving.

8. Learn

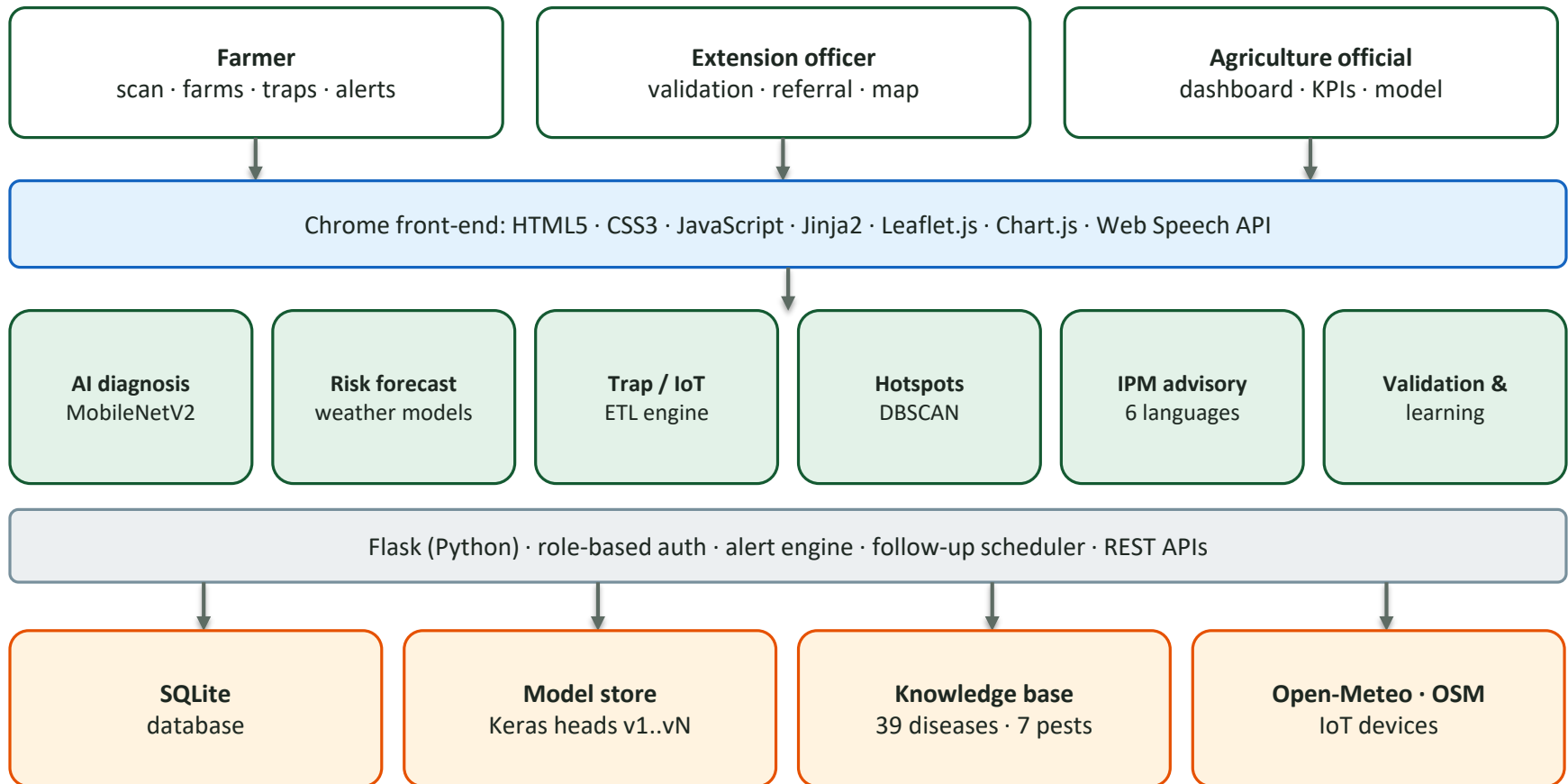
Retrain the model from field-confirmed images (continual learning).

9. Inform

Dashboards with surveillance KPIs for agriculture officials.

Methodology: Approach

Three-tier web system: browser front-end → Flask server with six intelligence modules → data & external services



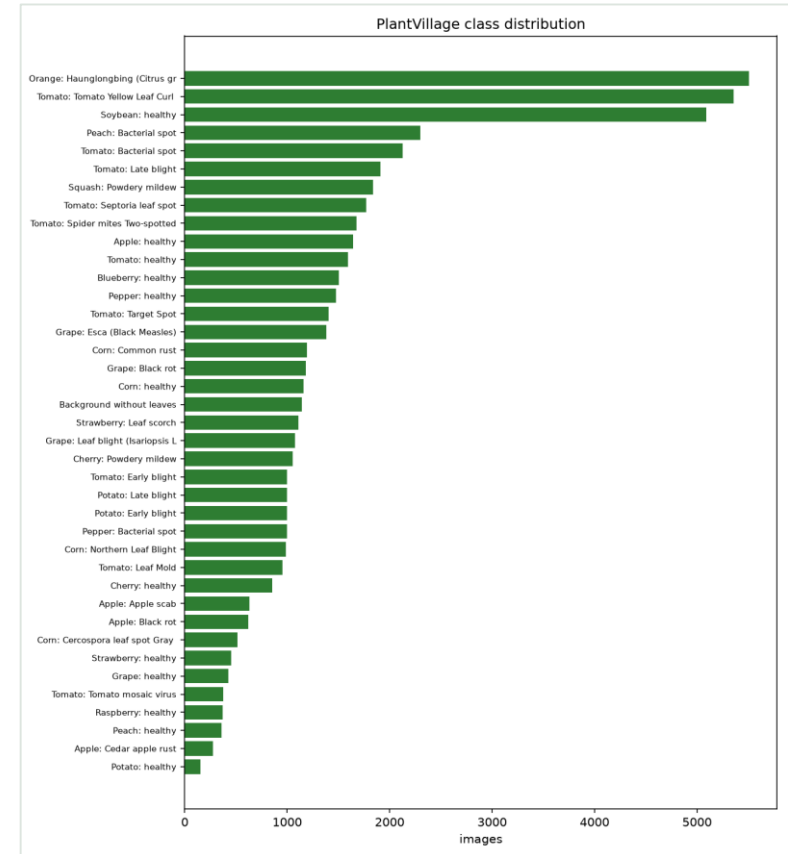
Approach: combine computer vision (what the farmer sees) with agro-meteorological risk models (what is likely next) and human expert validation (trust).

Methodology: Tools & Technologies

Layer	Technology	Purpose
Front-end	HTML5, CSS3, JavaScript, Jinja2	Responsive pages that run in Chrome (desktop & mobile)
Maps & charts	Leaflet.js + OpenStreetMap, Leaflet.heat, Chart.js	Hotspot map, heatmap, dashboards, weather chart
Back-end	Python 3.12, Flask 3	Routes, auth, REST APIs, alert & follow-up engine
Deep learning	TensorFlow 2.18 / Keras, MobileNetV2	Leaf image classification (transfer learning)
ML utilities	scikit-learn, NumPy, Pillow, Matplotlib	DBSCAN hotspots, metrics, image quality, plots
Database	SQLite	Users, farms, reports, traps, sensors, alerts
Weather	Open-Meteo forecast API (free)	Hourly temperature, humidity, rain for 10 days
Voice	Web Speech API (browser)	Read advisories aloud in the farmer's language
Hardware (sim.)	IoT weather station / smart trap via /api/sensor	Sensor & trap-count input

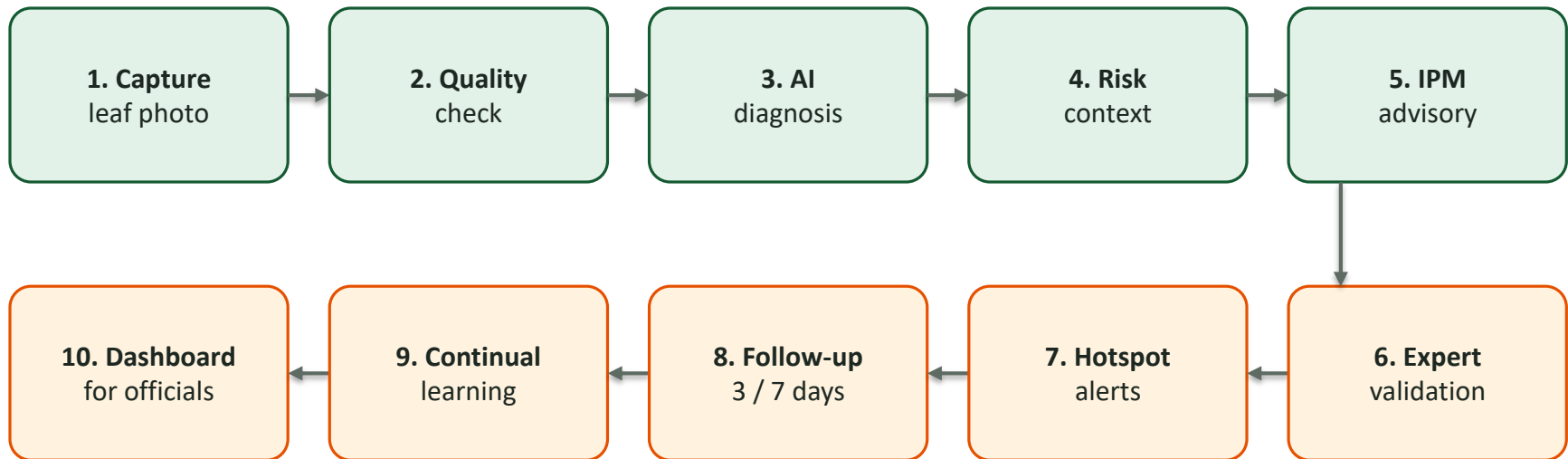
Methodology: Data Collection

- **Image dataset:** PlantVillage (Mendeley Data, without augmentation) — 55,448 leaf images, 39 classes (38 crop/disease classes of 14 crops + background).
- **Split:** stratified 70 / 15 / 15 → 38,813 train, 8,317 validation, 8,318 test images.
- **Weather:** Open-Meteo hourly forecast (past 3 + next 7 days) for each farm's GPS location.
- **Knowledge base:** 39 disease and 7 pest profiles — pathogen, symptoms, IPM actions, doses, pre-harvest intervals, ETLs — compiled from IPM package-of-practice literature.
- **Field data:** farm profiles, trap counts, sensor readings and reports entered through the app. Demo database: 107 farms, 425 reports (synthetic, for evaluation).



Process: Workflow

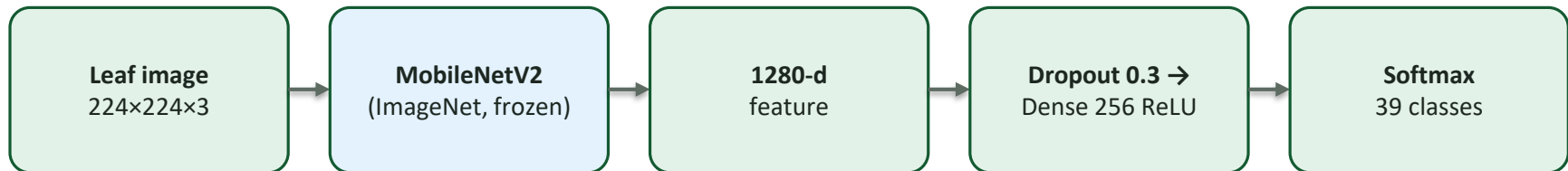
PROCESS



- **Farmer:** registers farm (GPS, crop, variety, sowing date, soil, irrigation) → gets daily risk alerts → scans a sick leaf → receives advice + audio.
- **System:** stores the image embedding, schedules a follow-up, refers to the nearest lab when confidence < 60% or disease is notifiable/viral.
- **Officer:** confirms or corrects the diagnosis and sets severity → farmer notified → hotspots recomputed → confirmed image added to training data.

Process: AI Diagnosis Model

PROCESS



- **Transfer learning:** ImageNet-pretrained MobileNetV2 is used as a frozen feature extractor — light enough for a CPU-only laptop and mobile deployment.
- **Two-stage training:** features of all 55,448 images are computed once (~40 min on CPU); the classification head then trains in 274 s.
- **Imbalance:** balanced class weights (classes range from 152 to 5,507 images); early stopping on validation accuracy; learning-rate decay on plateau.
- **Crop-aware inference:** if the farmer selects the crop, probabilities are re-normalised over that crop's classes; a strong mismatch is flagged.
- **Image quality gate:** Laplacian-variance blur check and exposure check ask the farmer to retake poor photos.
- **Continual learning:** each expert-confirmed image's embedding is stored; retraining fine-tunes the head (field samples ×5) and activates the new version only if held-out test accuracy does not drop > 0.5 pt.

Process: Risk, Hotspots & Advisory Logic

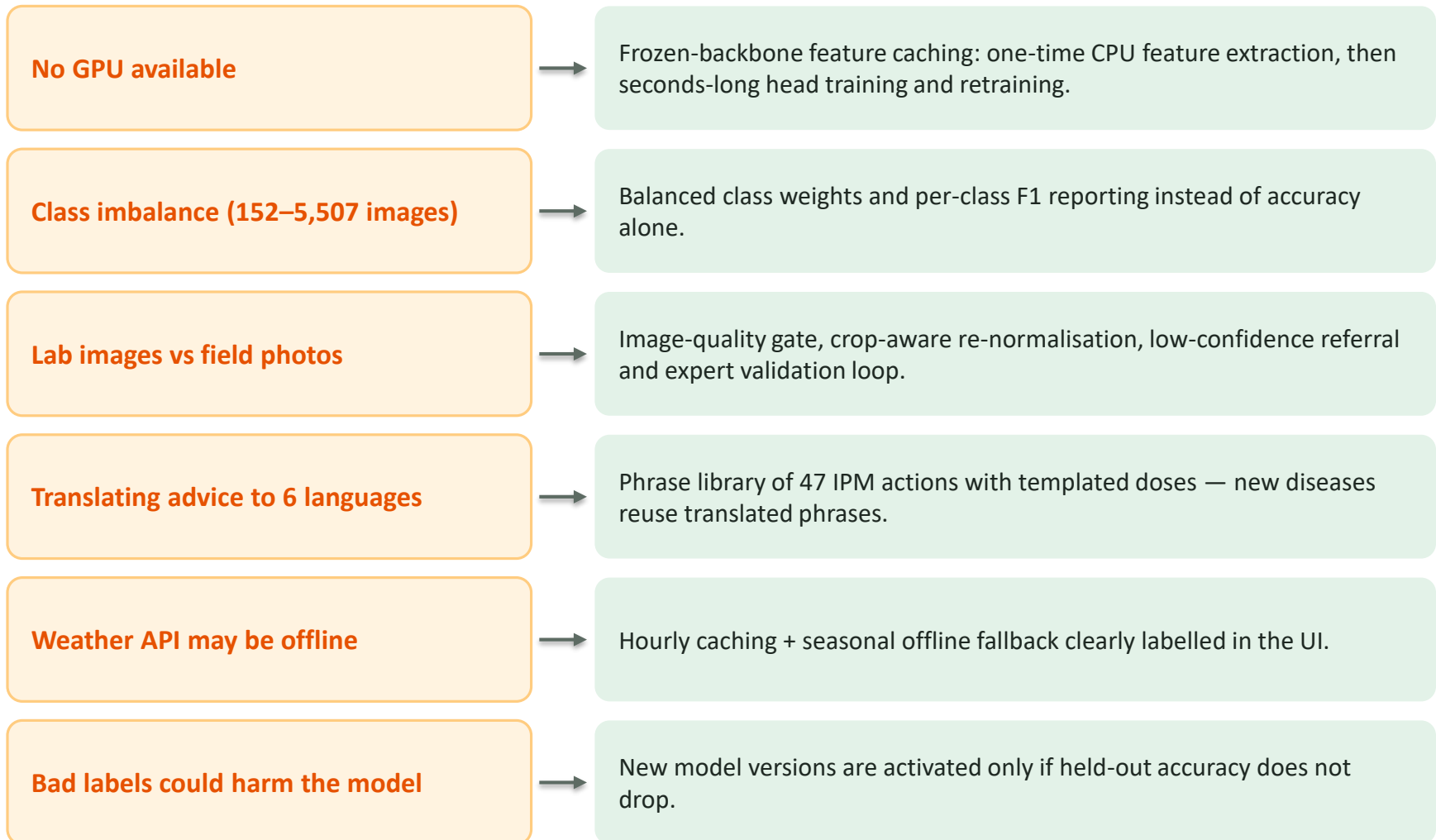
PROCESS

**Risk score = $85 \times \text{weather favourability} \times \text{crop-stage factor} \times \text{variety resistance} \times \text{drainage/irrigation factor}$
+ $6 \times \text{confirmed cases within 10 km in 21 days (max +25)}$ → Low / Moderate / High / Severe**

- **Weather models:** 12 disease models, e.g. late blight — Hutton criteria ($T_{\min} \geq 10^\circ\text{C}$ and $\geq 6\text{ h RH} \geq 90\%$); apple scab — Mills wet-hour table; spider mites — hot & dry spells.
- **Crop factors:** seedling 0.85 · vegetative 1.0 · flowering 1.15 · maturity 0.75; resistant variety 0.5, moderately resistant 0.75; poor drainage $\times 1.15$, sprinkler $\times 1.1$.
- **Pest traps & sensors:** counts normalised per trap per day/week and compared with ETL (e.g. pink bollworm 8 moths/trap/night, BPH 10/hill); IoT leaf-wetness raises humid-disease risk.
- **Hotspots:** DBSCAN with haversine distance (eps 5 km, ≥ 3 cases, last 30 days) per disease; farms of the same crop within radius + 5 km are alerted.
- **IPM advisory:** cultural → biological → chemical (last resort, dose + pre-harvest interval) → safety → referral → follow-up; composed from 47 translated action phrases.

Process: Challenges & Adaptations

PROCESS



Results: Model Performance

RESULTS

97.01%

Test accuracy

99.82%

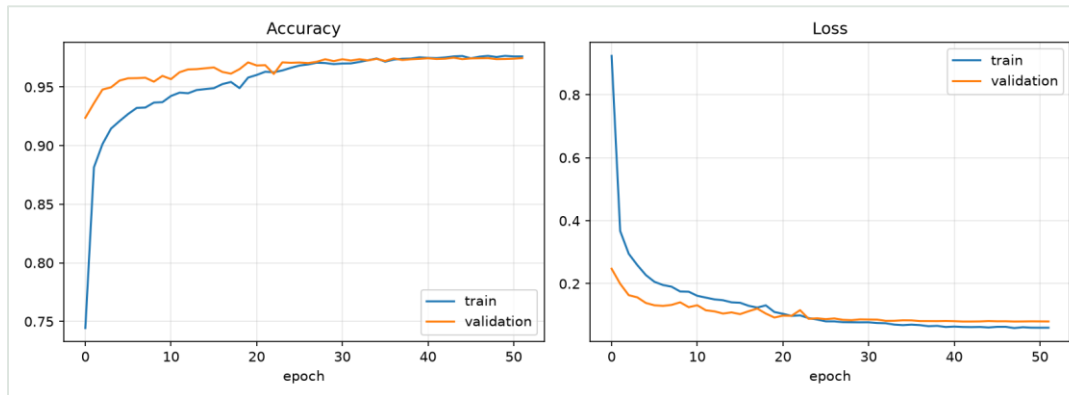
Top-3 accuracy

0.963

Macro F1-score

8,318

Unseen test images

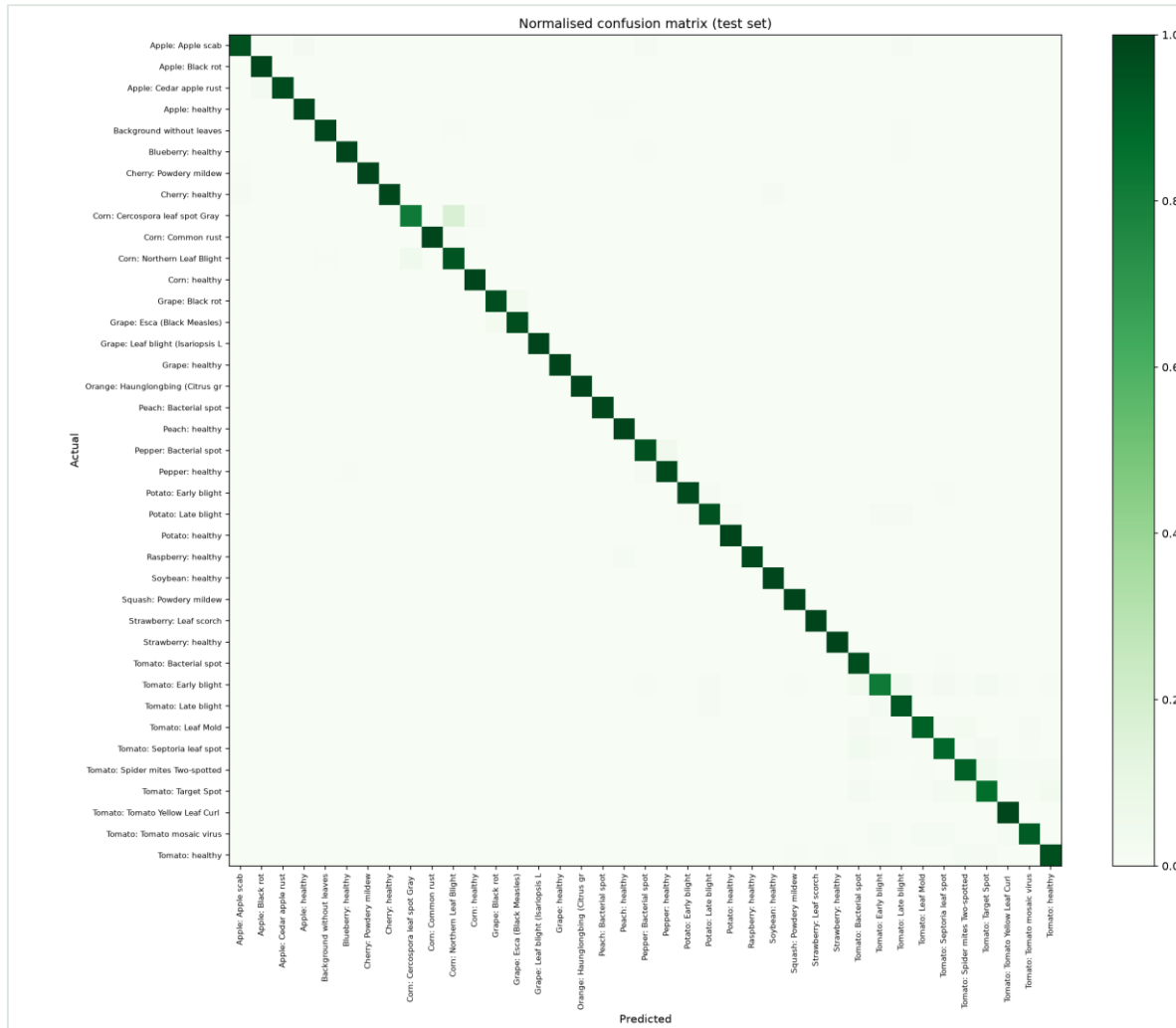


Weakest classes	F1
Tomato – Early blight	0.843
Maize – Gray leaf spot	0.857
Tomato – Target spot	0.867
Tomato – Mosaic virus	0.904
Tomato – Septoria leaf spot	0.915

- Validation curves track training curves closely — no over-fitting; early stopping after 52 epochs.
- Most errors are between visually similar leaf-spot diseases of the same crop (e.g. maize gray leaf spot vs northern leaf blight).
- Low-confidence predictions are routed to experts, so residual errors are caught before advice is finalised.

Results: Confusion Matrix

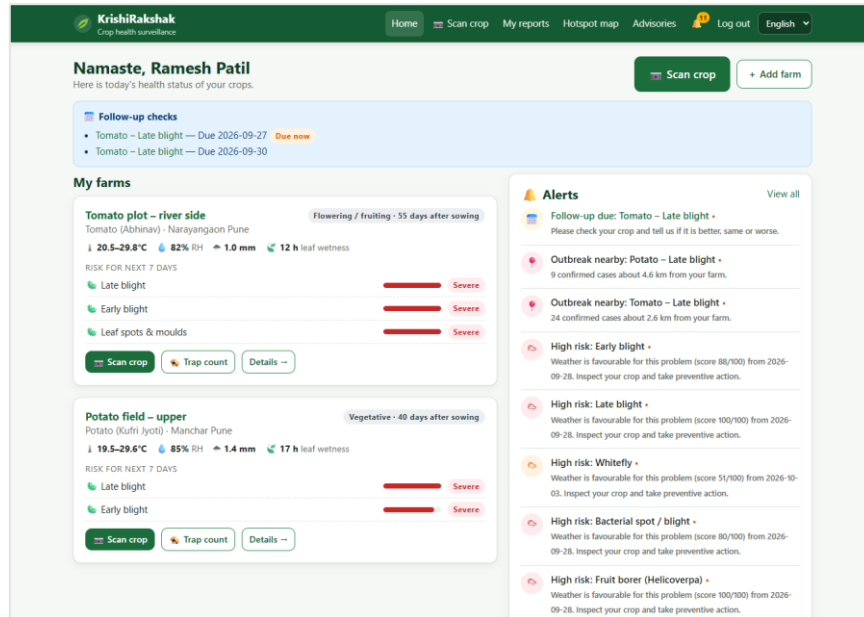
RESULTS



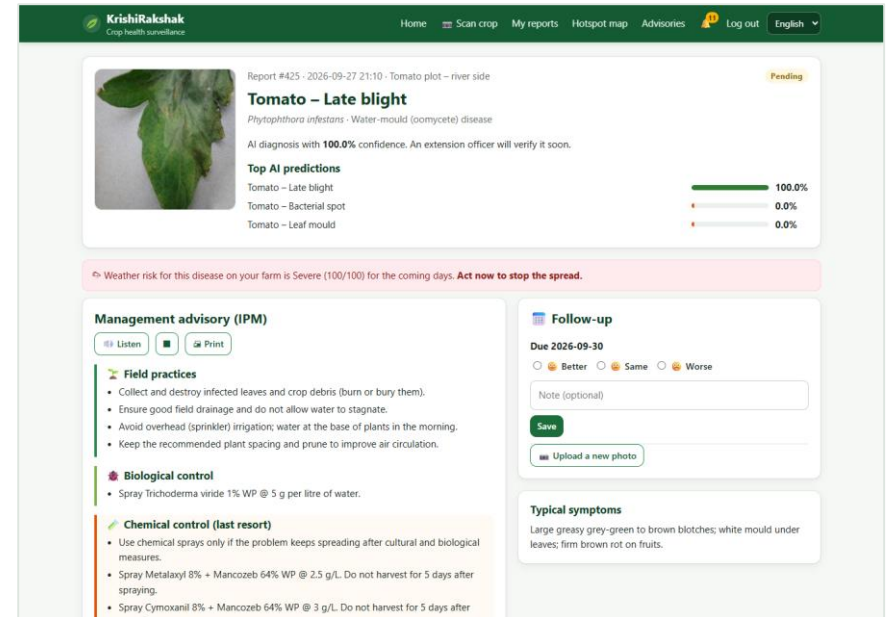
- **Strong diagonal:** almost every class is recognised correctly on unseen test images.
- **Background class:** non-leaf photos are detected and the farmer is asked to retake.
- **Confusions:** mainly within the same crop — the crop-aware mode and the top-3 list help the expert decide.
- **Benchmark:** Mohanty et al. (2016) reached 99.35% with fully fine-tuned GoogLeNet on PlantVillage; our frozen MobileNetV2 is far lighter and trains on a laptop CPU.

Results: Farmer Application

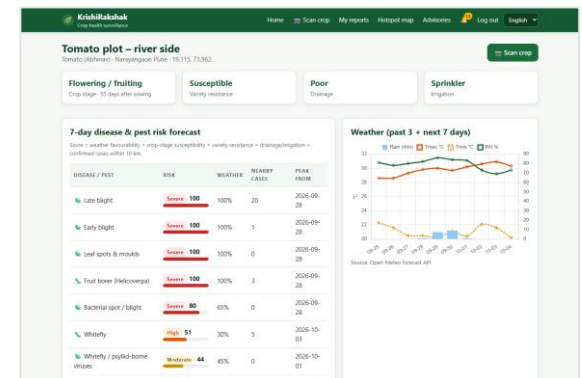
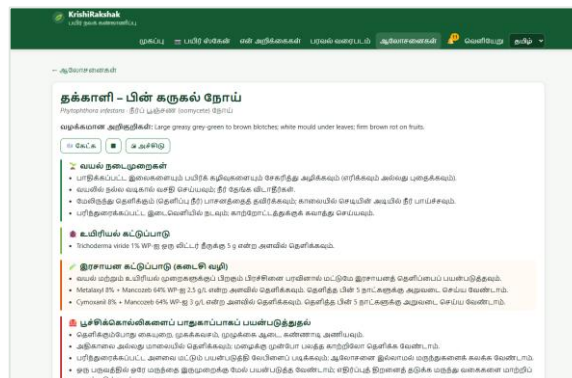
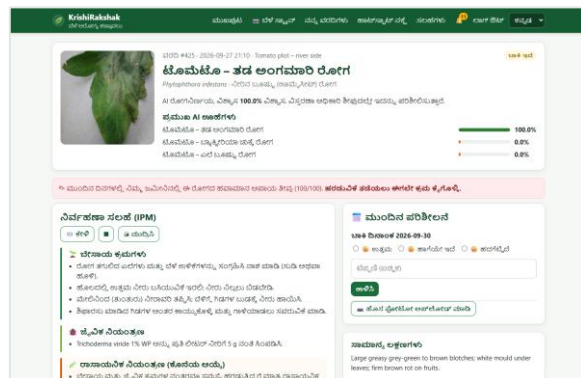
RESULTS



Farmer home: 7-day risk per farm, live weather, alerts

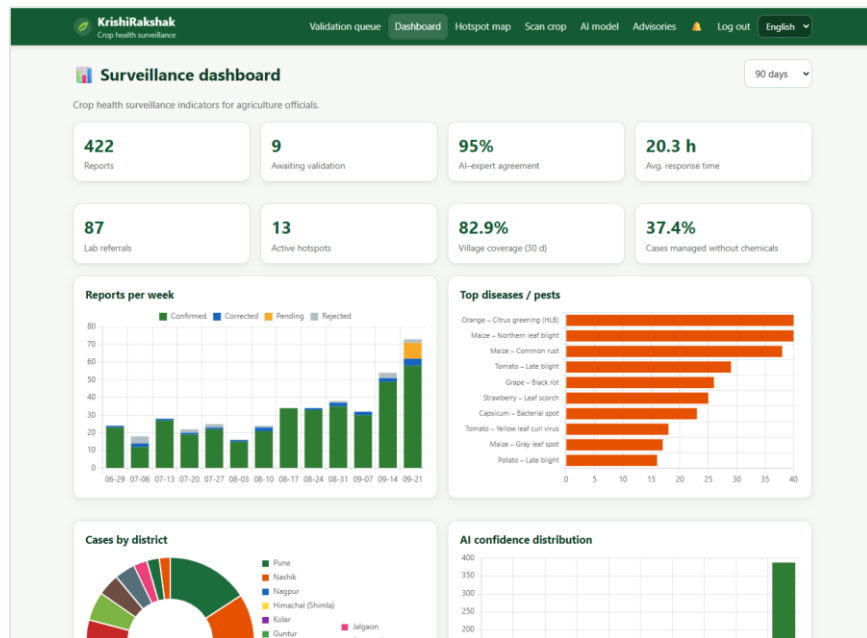


Diagnosis: top-3 predictions, weather link, IPM advisory

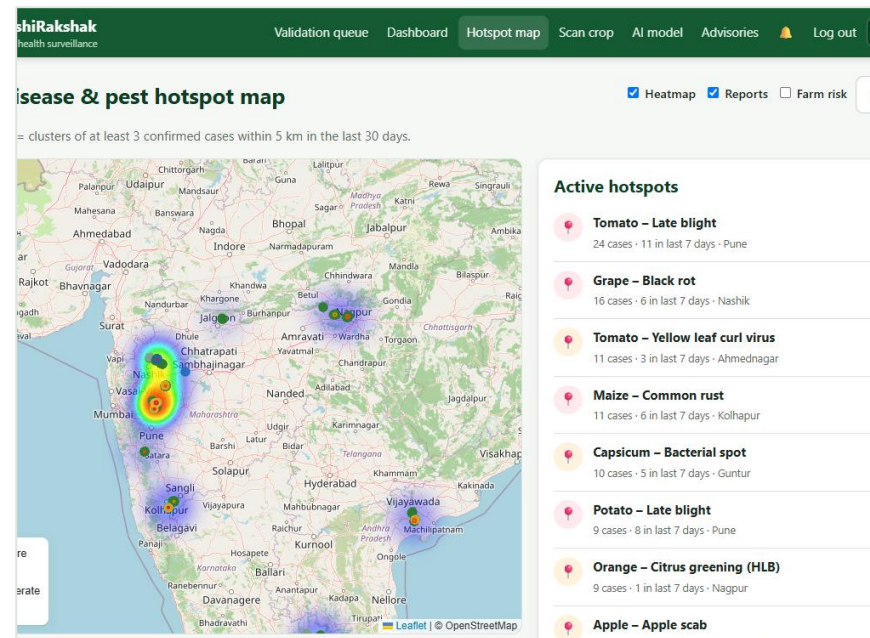


Results: Officer & Official Views

RESULTS



Surveillance dashboard (KPIs & charts)



Hotspot map: DBSCAN clusters + heatmap

95.0%

AI-expert agreement

20.3 h

avg. validation time

13

hotspots detected

37.4%

cases managed without chemicals

Measured on the demo database (425 reports over 90 days; AI predictions are real model outputs on held-out validation images, field context is simulated).

Results: Comparison with Existing Approaches

RESULTS

Capability	Manual extension	Photo-ID apps	KrishiRakshak
Image-based diagnosis	X (visit needed)	✓	✓ 38 classes + top-3
Weather risk forecast per farm	X	X / generic	✓ 12 models, 7 days
Crop stage / variety / soil	partly	X	✓
Pest traps & IoT sensors	manual	X	✓ ETL alerts + API
Hotspot mapping	X	X	✓ DBSCAN + heatmap
Expert validation loop	✓ (slow)	rare	✓ prioritised queue
Learns from field confirmations	—	X	✓ versioned retraining
Multilingual + audio advice	partly	partly	✓ 6 languages
Officials' dashboard	X	X	✓ KPIs & charts

Interpretation of results

- A lightweight CNN reaches 97.01% accuracy — enough for a first-line screening tool.
- Combining the photo diagnosis with weather risk tells the farmer not only what the problem is but how urgent it is.
- The expert loop turns AI into a decision-support tool, not a replacement for extension staff.
- IPM ordering and severity-based chemical advice support more targeted pesticide use.

Limitations

- PlantVillage images are lab photos on plain backgrounds; accuracy on real field photos will be lower until field data is collected.
- Only 14 crops / 38 classes; major Indian crops like rice, cotton and chilli need their own image data.
- Weather rules are simplified and not yet calibrated with local outbreak records.
- Demo field data is synthetic; IoT devices are simulated.
- Needs internet for live weather and map tiles.

Future Work & Conclusion

Future work

- Collect field images for rice, cotton, chilli, pulses; fine-tune the full network.
- Offline Android app with on-device TensorFlow Lite model.
- SMS / WhatsApp / IVR voice alerts for farmers without smartphones.
- Calibrate risk models with state pest-surveillance data; add soil-health cards.
- Real LoRa/GSM smart traps with automatic insect counting.
- Satellite (NDVI) crop-stress layers on the hotspot map.

Conclusion

KrishiRakshak delivers a complete crop-health system that runs in Chrome: image-based diagnosis, weather-based risk forecasting, pest-trap and sensor monitoring, hotspot mapping, expert validation, multilingual IPM advice, follow-up and continual learning, with dashboards for officials.

It supports the expected outcomes of the problem statement: earlier detection, reduced crop loss, more targeted pesticide use, faster extension response, wider surveillance coverage and better planning of preventive interventions.

References

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Thank you